Requirements Engineering and UML Use-Case Modelling

Community Lost and Found Matching Network

Problem Statement #59, Media Events and Community

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Lab 1 Submission

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1. Requirements Table

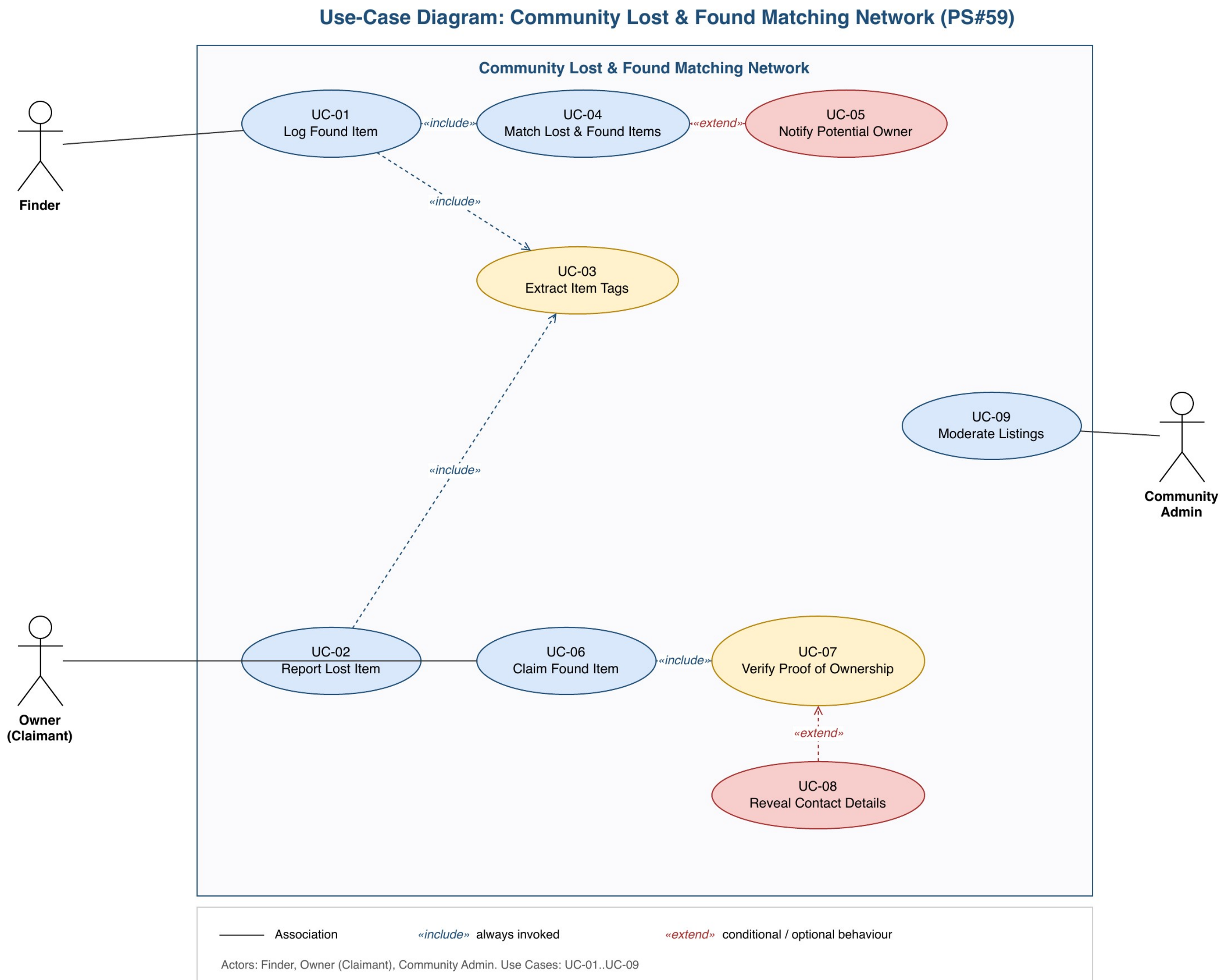
Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall match newly logged 'Found Item' posts with active 'Lost Item' reports based on category, color tags, and location radius.	High	Pass: Match probability calculated and notification dispatched to potential owner. Fail: Incompatible item categories matched.	Core value proposition of the platform: connects finders and owners automatically without manual browsing.
FR-002	Functional	The system shall extract visual tags (category, color, distinguishing features) from an uploaded item photo and text tags (keywords) from the item description at the time of posting.	High	Pass: Within 10s of image upload, at least 3 tags are auto-populated and shown to the poster for confirmation. Fail: No tags generated, or the poster must manually enter all tags.	Automated tagging is the required input for the FR-001 matching algorithm and reduces posting effort.
FR-003	Functional	The system shall require a claimant to correctly answer the proof-of-ownership question(s) set by the finder before a claim is approved.	High	Pass: Claim status changes to "Verified" only after all required questions are answered correctly. Fail: Claim is approved with an unanswered or incorrect question.	Prevents fraudulent claims and protects the finder from releasing an item to the wrong person.
FR-004	Functional	The system shall allow a Community Admin to flag, suspend, or remove any Lost or Found post reported as fraudulent, duplicate, or inappropriate.	Medium	Pass: A removed/suspended post no longer appears in search or matching results within 1 minute of the admin action. Fail: Post remains visible or matchable after removal.	Maintains trust and data quality across the community network.
FR-005	Functional	The system shall notify the Owner via push notification and email when a Found Item match with probability above the configured threshold is identified.	Medium	Pass: Notification delivered to the Owner within 2 minutes of a qualifying match. Fail: Owner is not notified for a match scoring above the threshold.	Timely notification increases the chance of successful reunification and item recovery.

Non-Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance and Security	Claimants must answer private security questions set by the finder before item handover contact details are revealed.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Protects the finder's personal contact information and prevents unauthorized claiming of items.
NFR-002	Reliability and Scalability	The matching engine shall complete geo-proximity matching for a newly logged item against all active listings within 5 minutes, sustaining at least 500 concurrent match requests without degradation.	Medium	Pass: Load testing confirms 95th-percentile match completion time under 5 minutes at 500 concurrent requests. Fail: Match completion exceeds 5 minutes or the system errors under simulated load.	Ensures the network remains responsive and useful as community adoption grows.

2. UML Use-Case Diagram



Built in draw.io. Actors: Finder, Owner (Claimant), Community Admin. Includes 4 <<include>> and 2 <<extend>> relationships across 9 use cases.

3. Use-Case Flow Specification

Community Lost and Found Matching Network, Problem Statement #59

Use Case	UC-06: Claim Found Item
Primary Actor	Owner (Claimant)
Supporting Actors	Finder, Community Admin (in alternate flow only)
Related Use Cases	Includes UC-07 (Verify Proof of Ownership); Extended by UC-08 (Reveal Contact Details); Follows UC-04/UC-05 (Match and Notify)

Preconditions

- The Owner has an active Lost Item report in the system.
- A Found Item post has been matched to that Lost Item report (UC-04) and the Owner has received a match notification (UC-05).
- The Finder has configured at least one proof-of-ownership question on the Found Item post.
- The Owner is authenticated in the system.

Postconditions

- **Success:** Claim status is set to “Verified: Awaiting Handover”; the Finder's and Owner's contact details are mutually revealed; both parties are notified.
- **Failure:** Claim status is set to “Rejected: Verification Failed”; no contact details are revealed; the attempt is logged for the Finder's and Admin's visibility.

Main Success Scenario

1. Owner selects “Claim This Item” on a matched Found Item post.
2. System displays the proof-of-ownership question(s) set by the Finder.
3. Owner submits answer(s) to the question(s).
4. System validates the submitted answer(s) against the Finder's stored answer(s) (includes UC-07: Verify Proof of Ownership).
5. System confirms all answers are correct.
6. System sets the claim status to “Verified” and triggers (extends UC-08: Reveal Contact Details).
7. System reveals the Finder's contact details to the Owner, and the Owner's contact details to the Finder.
8. System sends a confirmation notification to both Finder and Owner with suggested handover instructions.
9. Use case ends successfully.

Alternate Flow: 5a. Incorrect Answer(s)

Trigger: at step 5, one or more submitted answers do not match the Finder's stored answer(s).

1. 5a1. System displays an “Answer does not match” message and does not reveal any contact details.
2. 5a2. Owner is prompted to re-attempt the proof-of-ownership question(s).
3. 5a3. If answers are still incorrect after 3 total attempts, the claim is marked “Rejected: Verification Failed,” the attempt is logged, and the Finder is notified of a failed claim attempt.
4. 5a4. Owner may submit a manual dispute to Community Admin (UC-09: Moderate Listings) for human review.
5. Use case ends in failure (or resumes at step 2 if attempts remain).